Power BI: Importing & transforming a CSV with Power Query

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1. Getting started: connect & open in Power Query

1. Open Power BI Desktop and choose **Blank report** to start a new .pbix.

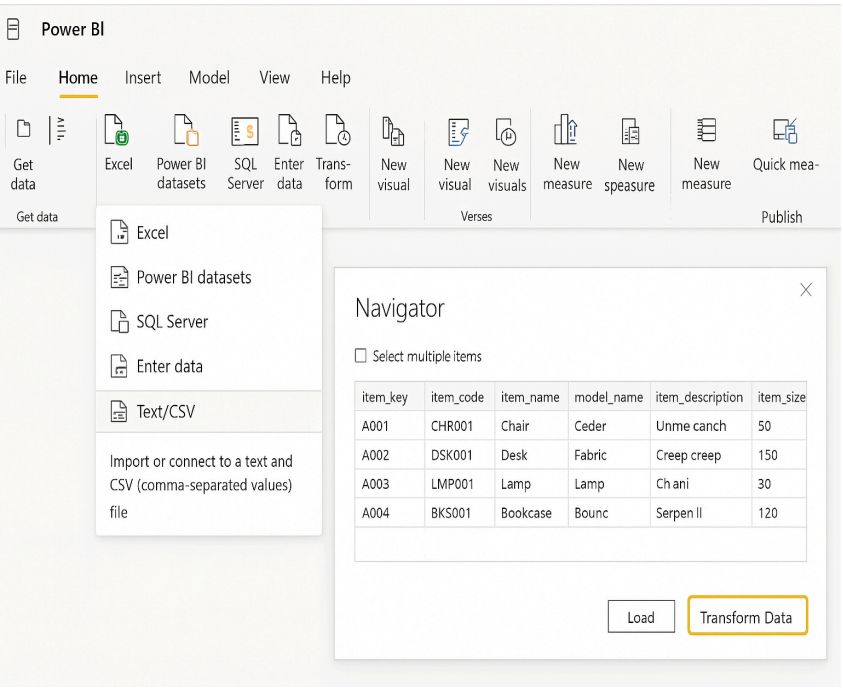
2. Use **Get data → Text/CSV** to import a CSV file (in the demo: list_of_products.csv). This opens a preview and kicks off a query to read the file.

    - **Tip / warning:** While Power BI is establishing the connection and running queries, *do not click around repeatedly*. Interrupting query processing can stall the app and might require restarting Power BI. Wait patiently until the preview and prompts finish.

3. When previewing the CSV you get a grid showing detected columns (example columns in demo: item_key, item_code, item_name, model_name, item_description, item_size, item_cost, item_price).

4. Instead of loading directly, click **Transform Data** to open the Power Query Editor when you plan to clean/prepare the data first.

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2. Saving the file & applying pending changes

1. In Power BI Desktop, save the workbook (e.g., products_transformation.pbix) as soon as possible. The instructor clicks **Save → Browse → Save As** and confirms the file name.

2. Power BI may show "pending changes" for the query steps — you can choose to **Apply** immediately or later. In the session the presenter chooses **Apply** and waits for the operations to complete.

    - The top-left window title shows the .pbix filename when saved.

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3. Power Query Editor — initial observations & auto-steps

When Power Query loads the CSV it commonly runs a few automatic steps:

- **Source** — the raw CSV load.
- **Promoted Headers** — first row promoted to column headers (if necessary).
- **Changed Type** — Power BI attempts to detect data types per column.

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Important note: Power Query's automatic **Changed Type** may not always be correct for your business use — it's the analyst's job to verify and correct types.

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4. Understanding data types & icons

Power Query uses small icons to show inferred column types:

- ABC — text (categorical)
- abc123 or 123 — whole number / integer
- 1.2 — decimal number

In the demo:

- item_key initially detected as numeric (1, 2, 3) but semantically it is an identifier (categorical). The instructor changes item_key from whole number to **Text** so it's treated as a categorical code and not usable in numeric calculations.

- item_code, item_name, model_name, item_description, item_size are categorical (ABC).
- item_cost, item_price are numeric (decimal) and left as decimal number type.

Best practice: Always verify inferred types and convert columns to the *most appropriate* type for their semantic meaning (ID → Text; Price/Cost → Decimal).

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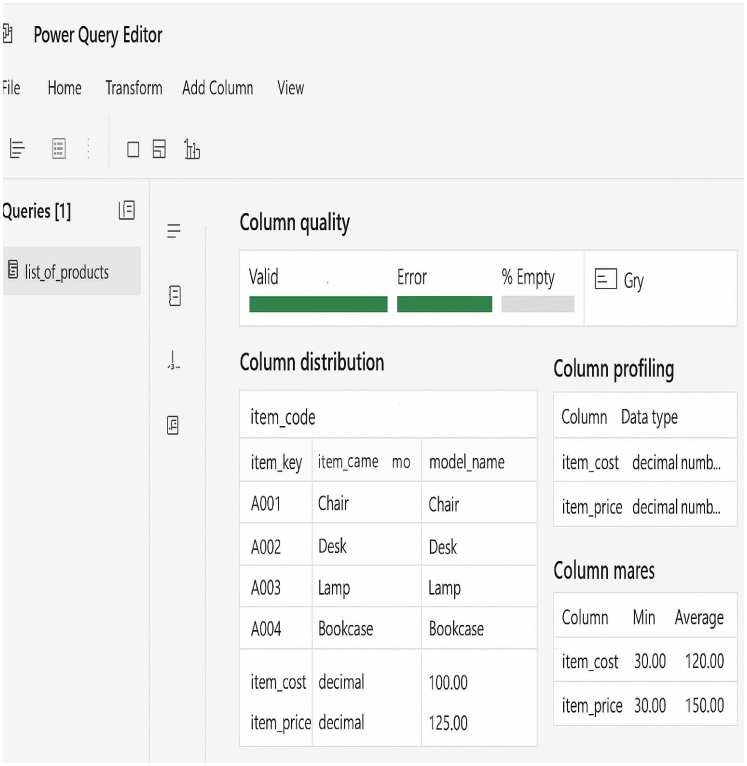
5. Column quality, distribution & profiling

Power Query offers in-built column diagnostics:

- **Column Quality** — shows % Valid, % Error, % Empty for each column. In the demo all columns show 100% valid, 0% error, 0% empty (clean file).
- **Column Distribution** — visualizes distinct values / frequency distribution for a column (useful for identifying outliers, duplicates, or skew).
- **Column Profile** — provides detailed statistics (min, max, average, standard deviation for numeric columns; distinct count, top values for categorical columns).
- **Column Statistics** — quick numeric statistics for numeric columns such as item_cost and item_price.

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Tip: Enable these (View → Column quality / distribution / profile) while cleaning to quickly identify problems or unusual values.



6. Requirement example: add a discounted price column

Scenario given: manager wants a new column that shows the price after a **10% discount** on the current item_price. Two mathematically equivalent methods are demonstrated.

Method A — Multiply by 0.9 (standard operation)

1. Select the item_price column.
2. Go to **Add Column → Standard → Multiply**.
3. Enter 0.9 as the multiplier and click **OK**.
4. Power Query adds a new column named (by default) Multiplication — rename it (double click header) to item_new_price_1.

- This produces the discounted price directly: item_price * 0.9.

Behavioral note: Because this operation used the standard numeric operation pathway, Power Query inferred the resulting column as a decimal (1.2) data type automatically.

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Method B — Subtract 10% using a Custom Column

1. Select **Add Column → Custom Column**.
2. Name the new column item_new_price_2.
3. Enter the formula:
   item_price - (0.1 * item_price)
   (or use the column selector to insert item_price, type - (0.1 *, insert item_price again, then close the bracket).
4. If syntax is correct, Power Query shows "No syntax errors have been detected" — click **OK**.

- This computes item_price - (item_price * 0.1) which equals item_price * 0.9.

Behavioral note: When created with Custom Column, Power Query sometimes assigns a different initial type — in the demo the custom column showed abc123 (mixed/unknown) and needed to be changed to Decimal Number (1.2) manually.

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7. Ensuring consistent numeric formatting (rounding / decimal places)

After producing the discounted columns, some values had varying decimal places (some 3 decimals, some 2). To enforce uniformity:

1. Ensure the column data type is Decimal Number (1.2). Right-click the column header → **Change Type → Decimal Number**.

2. With the column of correct type, use **Transform → Round → Round** (or **Round Up**, **Round Down** depending on policy).

3. When selecting **Round** specify the number of decimal places — in the demo the presenter used 2 decimal places and applied to both item_new_price_1 and item_new_price_2.

Key point: If a column has wrong type (e.g., detected as text/abc123), rounding or numeric transforms may not behave as expected — set the type first.

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8. Comparing the two methods — practical differences

- **Result:** Both methods return identical numeric results (mathematically equal).
- **Workflow difference:** The standard *Multiply* operator automatically inferred the output type as numeric; the *Custom Column* required manual type correction in the demo.
- **When to use which:**
    - Use standard transformations for quick numeric operations when available (Add Column → Standard → operation).
    - Use Custom Column when logic needs combining columns, using conditional logic, or applying a formula not available in standard menu options.

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9. Power Query tips & best practices highlighted

1. **Save early and frequently.** The presenter saved the .pbix before running many transformations.

2. **Avoid interrupting query processing.** Repeated clicks can hang Power BI during query evaluation.

3. **Verify column data types manually.** Don't blindly trust Power BI's automatic type detection — change IDs to Text and prices to Decimal as needed.

4. **Use Column Quality and Profile** to quickly assess missing values, errors, and distributional issues.

5. **Name columns clearly.** The demo renames generated column headers to meaningful names like item_new_price_1 and item_new_price_2.

6. **Round and format only after you confirm column types.** Rounding on a non-numeric type will fail or have unexpected behavior.

7. **Prefer built-in operations whenever possible** for clarity and better type inference; use custom formulas when operations require column references or more complex logic.

8. **Document transformations:** Power Query records steps in the Applied Steps pane — these are reproducible and form a clean ETL history.

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10. Example M expressions (for reference)

- Multiplication route (Power Query GUI creates equivalent M step behind scenes):
  = Table.AddColumn(#"Previous Step", "item_new_price_1", each [item_price] * 0.9, type number)

- Custom column formula demonstrated in the UI corresponds to:
  = Table.AddColumn(#"Previous Step", "item_new_price_2", each [item_price] - (0.1 * [item_price]))

(These are conceptual examples — Power Query generates its own naming and exact step references.)

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11. Final checks & applying changes

1. After cleaning and creating columns, check **Column Quality**, **Distribution**, and **Profile** again for the new columns.

2. Confirm data types and rounding are consistent.

3. Apply changes (if not already applied) so the data is loaded into the report model for visualization.

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12. Quick summary / takeaways

- Importing a CSV via **Get data → Text/CSV** and choosing **Transform Data** is the standard workflow when you want to clean before loading.
- Power Query's automatic steps are convenient but *must* be validated — particularly data types.
- Creating new calculated columns can be done through **Add Column → Standard** (fast numeric operations) or **Add Column → Custom Column** (flexible formulas). Both produced the same 10% discount column in the demo (multiply by 0.9 vs. subtract 10%).
- Keep numeric columns as Decimal Number for precision; use **Round** to enforce consistent formatting.
- Use built-in profiling tools to ensure data quality; avoid interrupting query processing to prevent hangs.
- Save often and maintain readable step names so transformations are auditable and reproducible.

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13. Suggested next steps (for practice)

1. Repeat the exercise but create a new discounted price with a variable discount stored in a separate table — join and apply using a lookup or parameter.
2. Try a conditional discount: e.g., if [item_cost] > 50 then [item_price] * 0.85 else [item_price] * 0.95 in a Custom Column.
3. Explore applying number formatting in the report view so the item_new_price shows currency symbols and thousand separators.
4. Save versions of the .pbix after major steps (e.g., before/after heavy transformations) to create restore points.